

Question 1.

from the question we can get $\Sigma = \{a, b\}$; $\mathcal{L} = \{w \in \Sigma^* \mid w \text{ begins with } b \text{ and ends with } a\}$

5-tuple of DFA.

1. $Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{a, b\}$

$\delta: Q \times \Sigma \rightarrow Q$

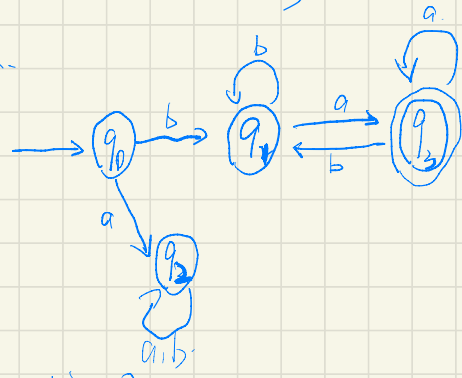
$q_0 \in Q$ is the initial state.

$q_2 \in Q$ is the accepting state.

6. transition function.

	a	b
q_0	q_2	q_1
q_1	q_3	q_1
q_2	q_2	q_2
q_3	q_3	q_1

2.



Question 2.

First

NFA.

$Q = \{q_1, q_2, q_3\}$

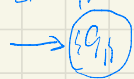
$\Sigma = \{0, 1\}$

$\delta = Q \times \Sigma^* \rightarrow P(Q)$

q_1 is initial state.

q_3 accepting state

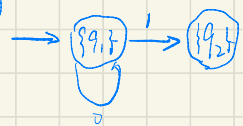
① start from q_1



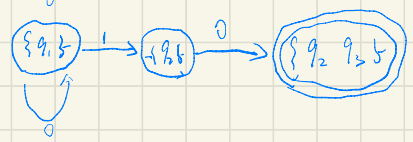
②



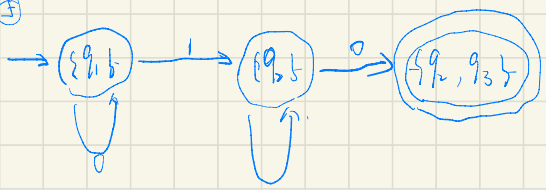
③



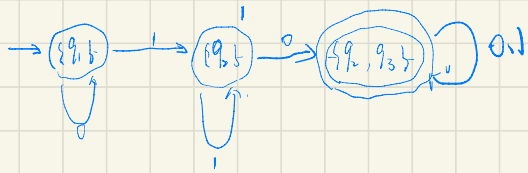
④



⑤



⑥



transition function

state	0	1
$\{q_1\}$	$\{q_1\}$	$\{q_2\}$
$\{q_2\}$	$\{q_2, q_3\}$	$\{q_2\}$
$\{q_2, q_3\}$	$\{q_2, q_3\}$	$\{q_2, q_3\}$

Question 3.

from the question we know that regular expression $(00)^*(11)$.

$0: \rightarrow 0 \rightarrow 0$

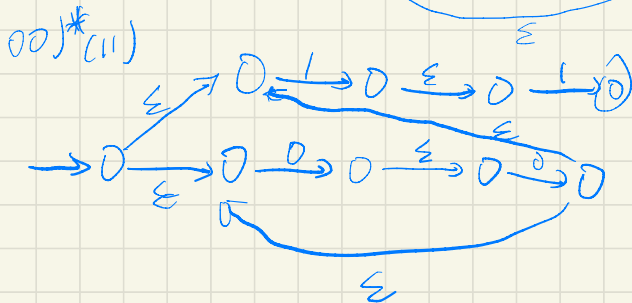
$1: \rightarrow 0 \rightarrow 1 \rightarrow 0$

$00: \rightarrow 0 \rightarrow 0 \rightarrow 0 \xrightarrow{\varepsilon} 0 \rightarrow 0$

$(11): \rightarrow 0 \rightarrow 1 \rightarrow 0 \xrightarrow{\varepsilon} 0 \rightarrow 1 \rightarrow 0$

$(01): \rightarrow 0 \rightarrow 0 \rightarrow 0 \xrightarrow{\varepsilon} 0 \rightarrow 1 \rightarrow 0$

$(00)^* \rightarrow 0 \xrightarrow{\varepsilon} 0 \rightarrow 0 \rightarrow 0 \xrightarrow{\varepsilon} 0 \rightarrow 0 \rightarrow 0$



Question 4.

proof:

1. assume that A_1 is regular

then, by pumping lemma for regular languages, there exists a pumping length $p \geq 1$ such that any string $s \in A_1$ with $|s| \geq p$ as

$$s = xyz$$

$$|y| \geq 1$$

$$|xy| \leq p$$

$$\forall i \geq 0 \quad xy^iz \in A_1$$

2. let $w = a^p$.

Then $s = www = a^{3p}$.

Clearly $s \in A$, and $|s| \geq 3p > p$.

3. since $|xy| \leq p$ both x and y lie entirely within the first a^p .

Therefore $y = a^k$ for some $k \geq 0$.

so we have that $x = a^m$ $z = a^j$.

$$y = a^k$$

$s = xy \geq$ implies $a^{3p} = a^m a^k a^j \Rightarrow m+k+j = 3p$.

4. property 1 implies $xyy \geq GA$ but

$$\begin{aligned} xyyz &= a^m a^k a^k a^j \\ &= a^{m+k+j+k} \\ &= a^{3p+k} \notin A. \end{aligned}$$

This is a contradiction

Conclusion: $A = \{www \mid w \in \{a, b\}^*\}$ is not regular.